

Homework 3

Task 1:

The data set is the same from the previous homeworks, so its shape is the same with 2,509 samples and 12,309 features. One thing to notice here, I split the dataset into training data(80%) and testing data(20%), and after that I made sure to move the target patients from the assignment requirements from the training set to the testing set. I've done this, because for Task 2 I have to apply SHAP on those 5 patient samples (so they need to be in the test data). This is because SHAP are always measured against a baseline, and the training data sets that reference point.

For this task I trained several models such as: Decision Tree, Random Forest, Gradient Boosting Machine, XGBoost, LightGBM and a Catboost. For each one, I tried tuning the hyperparameters so that the results were very accurate. After this, I aggregated all the metrics for each model into a stacked bar plot (Accuracy, Precision, Recall, F1 and AUC) as seen below in Figure 1.

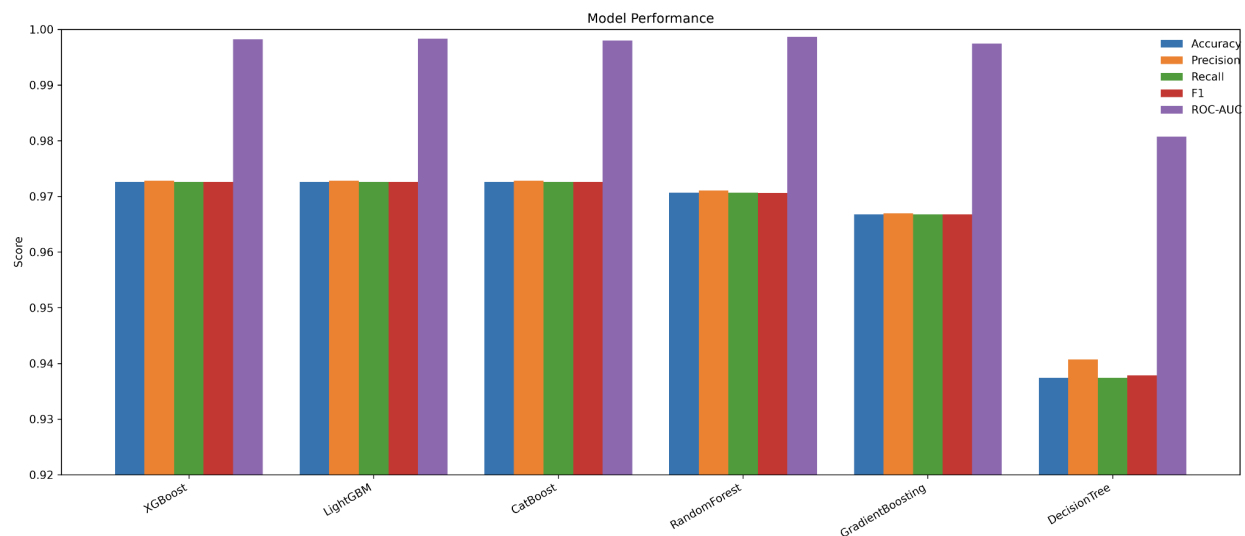


Figure 1. Model Performance

It can be observed that all Gradient Boosting based models, and also the Random Forest, performed really well, having the accuracies, precisions, recalls and f1s at around 0.97 and the AUC at 1.

For the next task, I will be choosing the LightGBM because it was the fastest model (GradientBoostingMachine and Catboost were by far the slowest, XGBoost was relatively fast, but was outperformed by LightGBM in terms of speed).

Task 2:

For this task I applied the SHAP algorithm to the training data and using those findings I extracted the SHAP values from the test data set. After that, I applied the SHAP for the five target patients from each one of the cancer classes to visualize on a force plot how well the patient was identified to belong to its correct class and what features contributed to that prediction, also which features contributed to not classifying that patient to the wrong class.

SHAP, also known as SHapley Additive exPlanations, is an algorithm that measures the individual contribution of each feature towards the final predicted result.

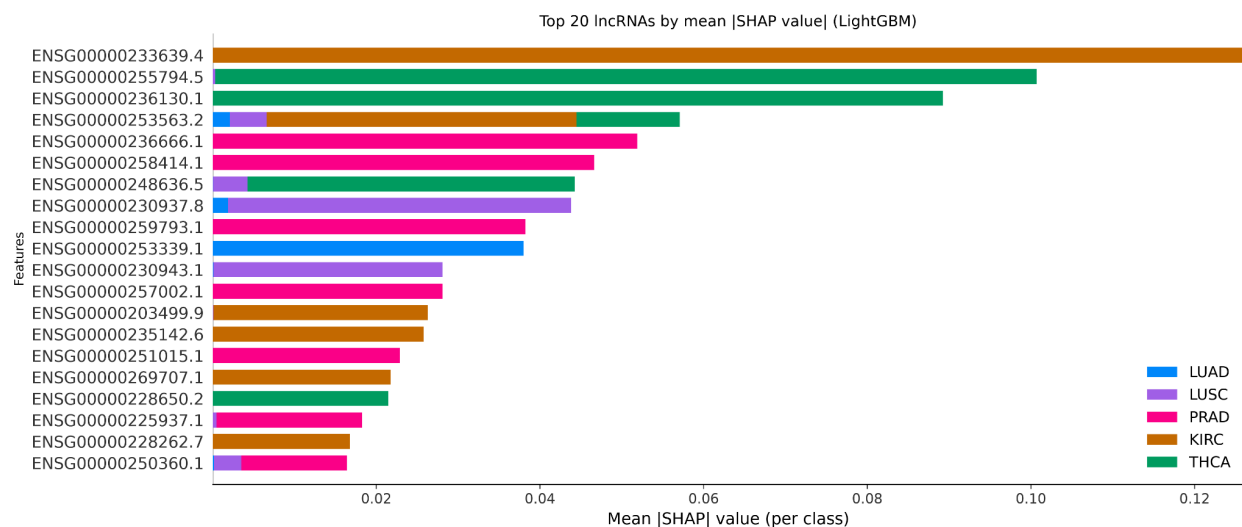


Figure 2. Top 20 features by mean |SHAP| value

Figure 2 presents the top 20 features of LightGBM model ordered by their mean SHAP value. It can be seen that most of the top features belong to exactly one type of cancer, for example, ENSG00000233639.4 matters only for KIRC and is the strongest feature overall, ENSG00000236130.1 is a THCA marker, and ENSG00000236666.1 is a PRAD marker. Only a few features are shared across classes such as: ENSG00000253563.2 contributes to KIRC, THCA and slightly to both lung types. ENSG00000248636.5 and ENSG00000250360.1 also mix two classes. LUAD has only one strong marker here (ENSG00000253339.1). That suggests the model has fewer strong lncRNA signals for separating LUAD from LUSC, which fits these two lung cancers being more similar to each other than to the other classes.

The X-axis displays the average size of a feature's effect on the prediction, ignoring whether the effect is pushed up or down. This tells us how much the model relies on that feature for that specific cancer class. The bigger the value the bigger the influence it has in predicting that class.

After this I plotted the top 10 features for each cancer type as seen in the below figures: 3, 4, 5, 6 and 7.

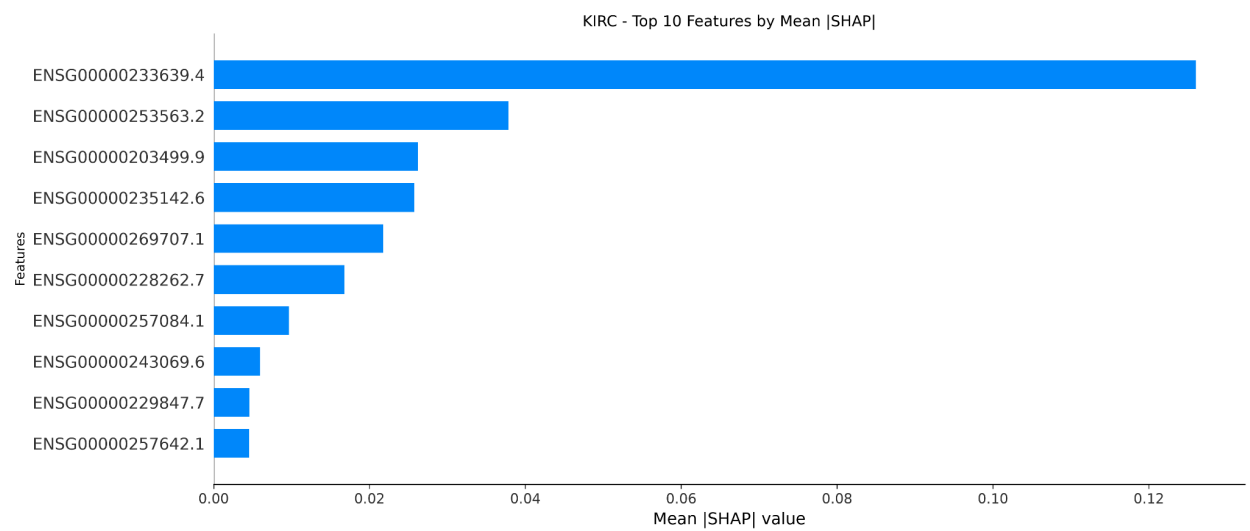


Figure 3. Top 10 features by SHAP values for KIRC class

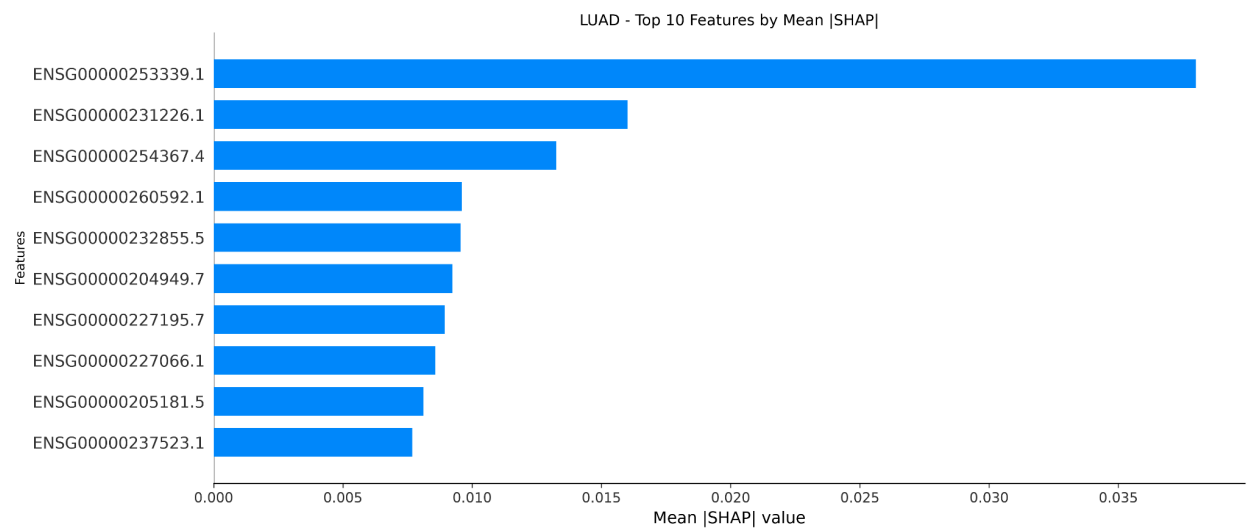


Figure 4. Top 10 features by SHAP values for LUAD class

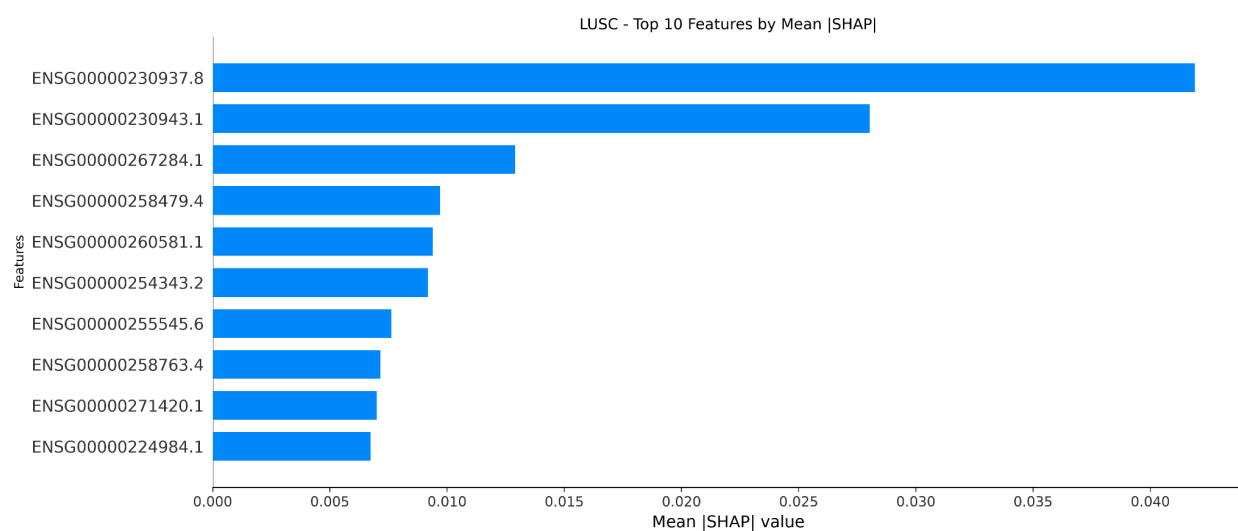


Figure 5. Top 10 features by SHAP values for LUSC class

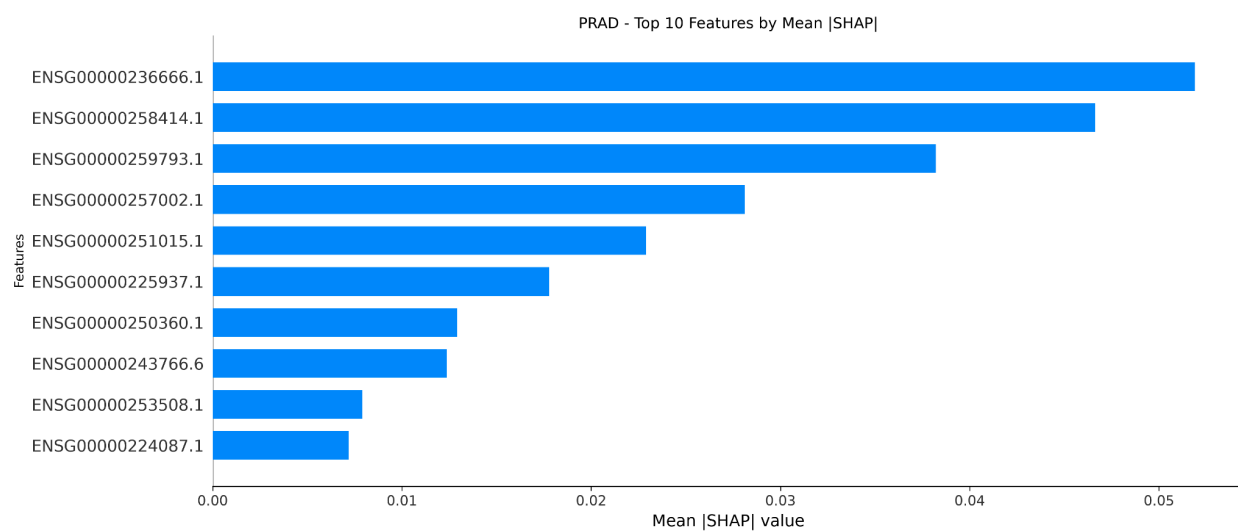


Figure 6. Top 10 features by SHAP values for PRAD class

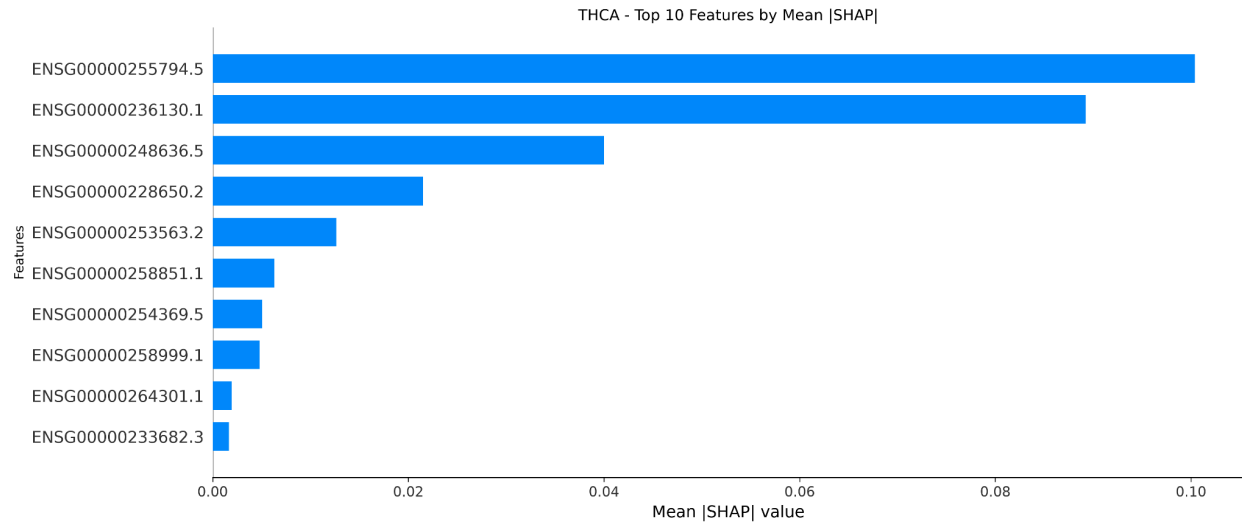


Figure 7. Top 10 features by SHAP values for THCA class

Figure 7, for example, shows the top 10 lncRNA features the model relies on to decide whether a sample is thyroid cancer (THCA). Two lncRNAs do most of the work. ENSG00000255794.5 (about 0.10) and ENSG00000236130.1 (about 0.09) are far ahead of the rest. On average, each one shifts a sample's THCA prediction by about 0.09–0.10. Two supporting features come next. ENSG00000248636.5 (about 0.04) and ENSG00000228650.2 (about 0.02) contribute noticeably less. Everything after the 5th feature is close to zero. The model barely uses those lncRNAs for THCA. So THCA has a compact signature: the model identifies it mainly from 2–4 lncRNAs. That usually means THCA is easy for the model to separate from the other cancer types. This fits thyroid tissue being quite different from lung, kidney and prostate.

The same explanations is applied to the other Figures: 3, 4, 5 and 6, the only thing that changes is the class being identified and of course, the specific genes that contribute the most towards that model prediction.

After displaying the bar charts for the top features, both for the global dataset and for each type of cancer, I computed the SHAP values for the 5 target patients given, one for each cancer class. I displayed the force plots to visualize better how some specific lncRNA genes are pushing the prediction towards a specific class.

As shown below in Figure 8, the plot for patient TCGA-CJ-4641-01A, we can see several things such as:

- Base value: the model's average output before it looks at this patient; roughly between 0.16 and 0.25, which is about the share of each class in the training data
- $f(x)$: model's final output for this patient
- Red bars: genes whose values push the score up
- Blue bars: genes whose values push the score down
- Labels: each label is a gene id

Force plots for KIRC patient TCGA-CJ-4641-01A

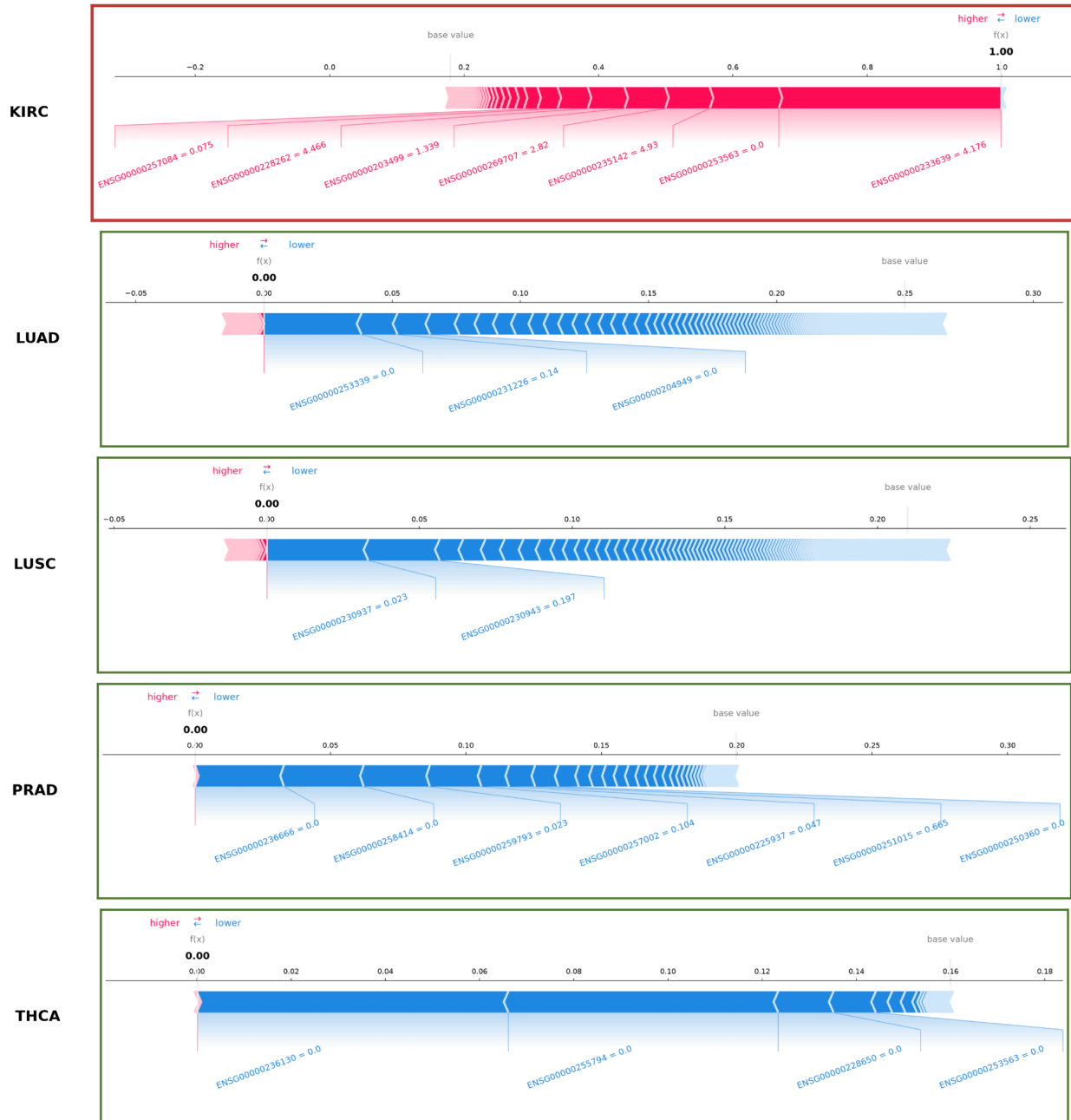


Figure 8. Force plots for KIRC patient in regard to every cancer type

The force plot tells us that:

- KIRC class (red bounded, the correct class for the given patient): the score rises from 0.18 to 1, with every visible contribution being red. The strongest come from ENSG00000233639 (4.176), ENSG00000253563 (0.0) and ENSG00000235142 (4.93), plus many smaller ones. The model is fully confident this is kidney cancer. It is easily explainable that the higher the number (gene's contribution) the better it pushes the

model's output towards the correct class. However, this gene ENSG00000253563 (0.0), has 0 score, which at first sight can seem a little bit off, perhaps there is a problem with the model, which is not the case. The model learnt that this gene is solely expressed in other types of cancer, so finding it here, eliminates that cancer class and pushes the score up towards the correct one.

- LUAD, LUSC, PRAD, THCA (the wrong classes): each score is pushed from its base value towards 0 by the blue contributions, so the model firmly rules these classes out. In lung cancers the “pushing” of the score down is down by many genes, each with small effect, while in PRAD and THCA there are a couple of genes that contribute the most to the predictions.

Down below there are the force plots for the other 4 patients in regard to every cancer type. The model is doing a great job as for each patient(given as test data, not previously learned), the score for the correct class is 1, and for the rest is 0.

Force plots for LUAD patient TCGA-99-8033-01A

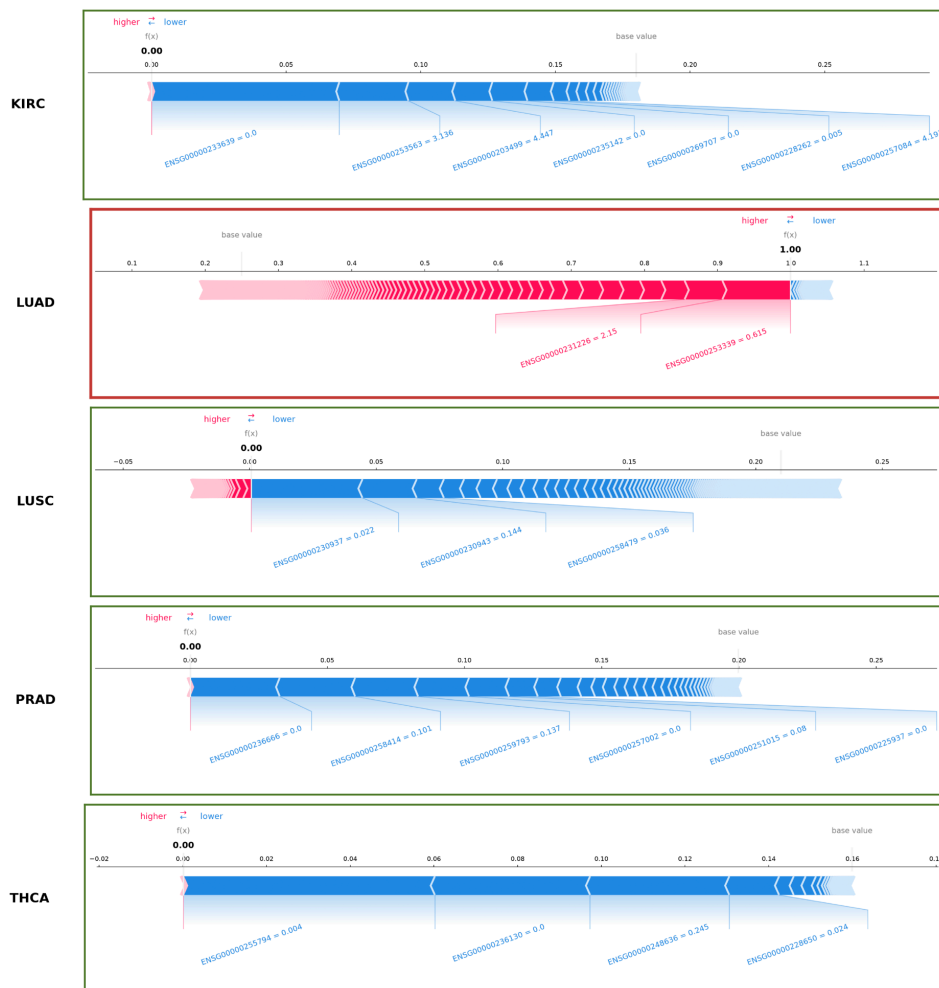


Figure 9. Force plots for LUAD patient in regard to every cancer type

Force plots for LUSC patient TCGA-66-2771-01A

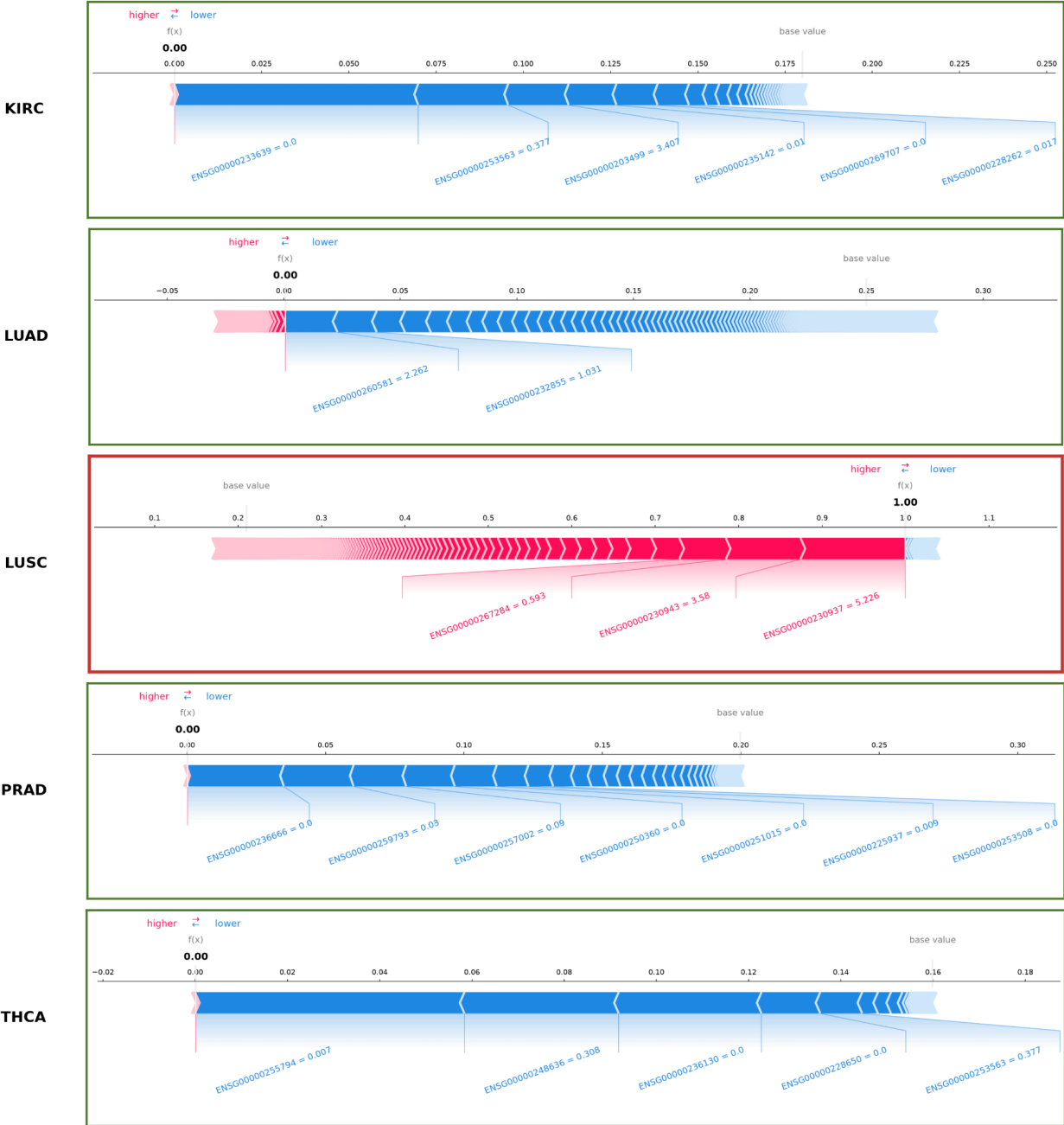


Figure 10. Force plots for LUSC patient in regard to every cancer type

Force plots for PRAD patient TCGA-HC-7821-01A

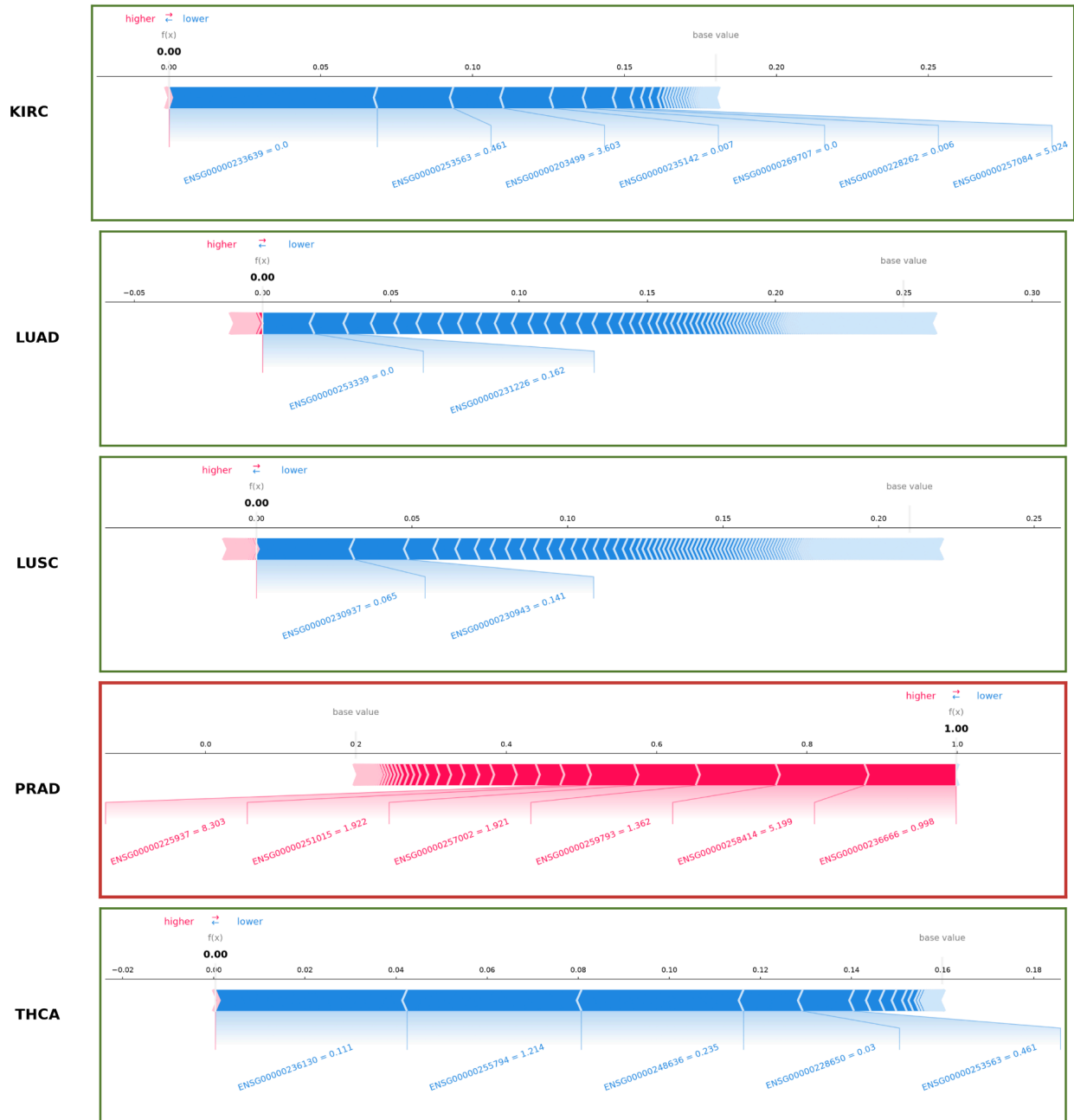


Figure 11. Force plots for PRAD patient in regard to every cancer type

Force plots for THCA patient TCGA-E3-A3DY-01A

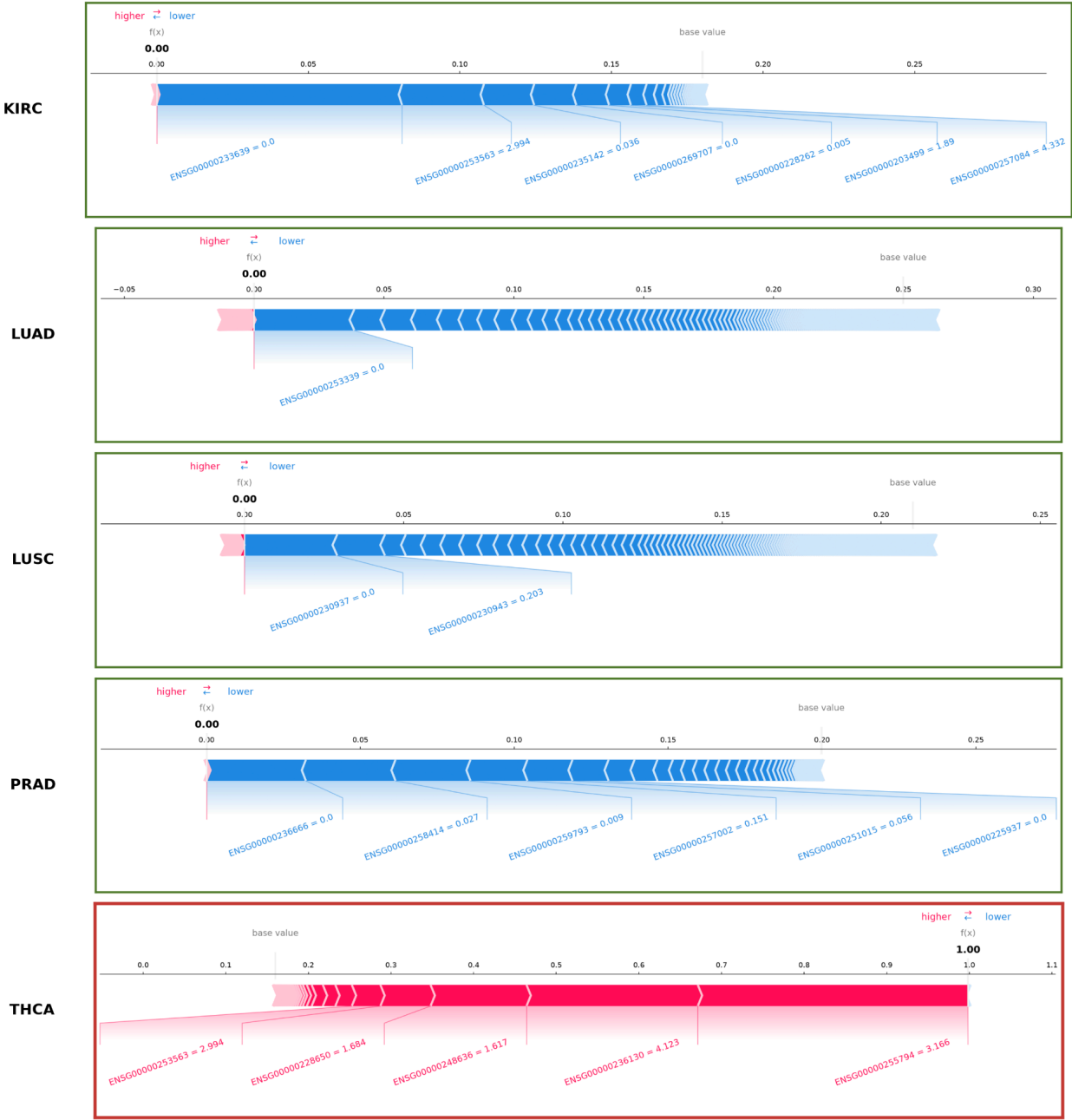


Figure 12. Force plots for THCA patient in regard to every cancer type